

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40023
Subject Title	Materials Modelling by Density Functional Theory
Credit Value	3
Level	4
Pre-requisite	AP20006
Objectives	Density functional theory (DFT) is a quantum-mechanical atomistic simulation method widely used in physics, chemistry, and materials science. This course aims to give the students insight into the basic concepts of DFT and applications of DFT in studying semiconductor materials. The emphasis of this course will be on both theoretical principles of DFT and hands-on experience in performing DFT calculations on simple solids using the DFT package, Quantum Espresso.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Appreciate the many-body theory of electronic system; - Explain the Hohenberg-Kohn theorems and Kohn-Sham equations; - Recognise the differences between various exchange-correlation functionals; - Describe the pseudopotential theory and construction of pseudopotentials. - Perform DFT calculations on the electronic properties of semiconductors.
Subject Synopsis/ Indicative Syllabus	Background of DFT Introduction and overview of DFT; Many-body system; Difficulties in solving many-electron problems Foundation of DFT Hartree approximation; Hartree-Fock approximation; Hohenberg-Kohn theorems Kohn-Sham Equations Effective exact single particle method; Exchange and correlation energies; Kohn-Sham equations Approximation to Functionals Local density approximation (LDA); Generalized gradient approximation (GGA); Non-local approximation; Hybrid functionals Pseudopotential Theory Norm-conserving pseudopotential; Construction of pseudopotentials Practical DFT calculations

	Introduction to Quantum Espresso; Workflow of performing DFT calculations																																													
Teaching/Learning Methodology	Lecture: The fundamentals of density functional theory will be introduced and explained in lectures. Examples will be given to illustrate the concepts in the lecture. Students are encouraged to participate in the in-class activities to enhance their knowledge acquisition and conceptual understanding. Tutorial: Students will practice and present topics of DFT applications in physics, providing them opportunities to reflect and disseminate the knowledge gained in the lectures. Computer laboratory: Students will conduct DFT calculations on real problems, which is essential for them to relate the basic concepts to practical applications.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr> </thead> <tbody> <tr> <td>1. Assignments</td><td>40 %</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr> <tr> <td>2. Oral Presentation</td><td>20 %</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr> <tr> <td>3. Computer Laboratory Report & Viva</td><td>40 %</td><td></td><td></td><td></td><td>√</td><td>√</td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="5"></td></tr> </tbody> </table> The assignments, including take-home question sets and in-class quizzes, are used to check students' learning progress and assess their understanding of basic concepts. The oral presentation is used to engage students on literature and information search of related topics. The computer laboratory reports and viva are used to evaluate students' skills of solving the problems and analysing the results.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignments	40 %	√	√	√	√		2. Oral Presentation	20 %	√	√	√	√		3. Computer Laboratory Report & Viva	40 %				√	√	Total	100 %					
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Student Study Effort Expected	Class contact:																																													
	▪ Lectures & Tutorials					26 Hrs.																																								
	▪ Computer Laboratory					18 Hrs.																																								
	Other student study effort:																																													
	▪ Self-study					76 Hrs.																																								
	Total student study effort					120 Hrs.																																								

Reading List and References	• Kieron Burke, The ABC of DFT, http://dft.uci.edu/doc/g1.pdf. • Giustino, F. Materials Modelling Using Density Functional Theory: Properties and Predictions. (Oxford University Press, 2014). • Martin, R. M. Electronic Structure: Basic Theory and Practical Methods. (Cambridge University Press, 2004). • Kohanoff, J. Electronic Structure Calculations for Solids and Molecules: Theory and Computational Methods. (Cambridge University Press, 2006). • Quantum Espresso, https://www.quantum-espresso.org/
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